

APPENDIX A: *Note- Tables contain formulary medications only***Table 1: Medications that cannot be crushed/opened.**

Generic (Active Ingredient)	Drug Brand Name	Administration Comments
Abacavir and abacavir containing combination products	Ziagen Epzicom	Do not crush or open. Liquid formulation available for abacavir.
alendronate	Fosamax	Do not chew or crush. Mucous membrane irritant.
aspirin enteric coated	Ecotrin	Do not chew or crush. Enteric coated. Slow release. Note: Utilize other formulations.
aspirin-dipyridamole	Aggrenox	Do not chew or crush. Slow-release capsule.
benzonatate	Tessalon Perles	Do not chew or crush; swallow capsule whole. Local mucosal irritant and anesthetic.
bisacodyl	Dulcolax	Do not chew or crush. Enteric coated; liquid filled.
bupropion	Wellbutrin	Do not chew or crush.
bupropion	Wellbutrin SR, Wellbutrin XL	Do not chew or crush. Slow release.
carbamazepine	Tegretol-XR	Do not chew or crush. Slow release. Teratogenic potential.
cobicistat	Tybost	Do not crush. Cobicistat is insoluble in water.
dabigatran	Pradaxa	Do not chew or crush. Note: breaking, chewing, or emptying contents of the capsule can result in increased exposure of the drug.
didanosine	Videx EC	Do not chew or crush. Enteric-coated, Slow release.
diltiazem SR	Cardizem CD	Do not chew or crush. Extended-release capsule.
disopyramide	Norpace CR	Do not chew or crush. Slow release.
divalproex	Depakote	Do not chew or crush tablet. Delayed or extended release. Teratogenic potential.
doxylamine/pyridoxine	Diclegis	Do not chew or crush. Slow release.
efavirenz/emtricitabine/tenofovir	Atripla	Do not chew or crush
emtricitabine/rilpivirine/tenofovir	Complera	Do not chew or crush
Elvitegravir, cobicistat, emtricitabine, tenofovir	Genvoya Stribild	Do not chew or crush
erythromycin	Ery-tab	Do not chew or crush. Enteric-coated.
fosamprenavir	Lexiva	Do not chew or crush
gefitinib	Iressa	Do not chew or crush. Note: Tablet may be dissolved in 4-8 ounces of water
guaifenesin	Mucinex ER	Do not chew or crush. Slow release.
guaifenesin/ dextromethorphan	Mucinex DM	Do not chew or crush. Slow release.
hyoscyamine	Levbid	Do not chew or crush. Note: Tablet is scored and may be broken in half without affecting release characteristics.
ibuprofen	Motrin	Do not chew or crush. Note: Crushing tablets has resulted in decreased bioavailability of drug.
isosorbide mononitrate	Imdur	Do not chew or crush Note: 30mg tablet may not be split
lenalidomide	Revlimid	Do not chew or crush Note: Teratogenic potential; health care workers should

		avoid contact with capsule contents.
levodopa/carbidopa	Sinemet CR	Do not chew or crush. Slow release. Note: Tablet is scored and may be broken in half if needed.
lithium	Lithobid	Do not chew or crush. Slow release.
lopinavir-ritonavir	Kaletra	Do not chew or crush. Film-coated.
maraviroc	Selzentry	Do not crush
mesalamine	Delzicol	Do not chew or crush. Slow release.
methylphenidate	Concerta	Do not chew or crush. Slow-release tablet.
metoprolol succinate	Toprol XL	Do not chew or crush. Note: Tablet is scored and may be broken in half.
mirabegron	Myrbetriq	Do not chew, crush, or divide tablets.
morphine	MS Contin	Do not chew or crush. Note: crushing, chewing, or dissolving tablets can cause rapid release and absorption of a potentially fatal dose
mycophenolate	CellCept	Do not chew, crush, or open. Teratogenic potential
mycophenolic acid	Myfortic	Do not chew or crush. Slow release; Enteric coated.
Naproxen EC	Naprosyn EC	Do not chew or crush. Slow release.
nevirapine	Viramune XR	Do not chew or crush. Slow release, Carcinogen.
niacin	Niacin ER	Do not chew or crush. Slow release.
Nifedipine	Procardia	Do not chew or crush. Note: pharmacokinetics may be altered if not taken whole.
Nifedipine	Procardia XL	Do not chew or crush. Extended release.
nitroglycerin	Nitrostat	Do not chew or crush. Note: Tablets are made to disintegrate under the tongue.
omega-3-acid ethyl esters	Lovaza	Do not chew or crush. Content of capsule may be erosive.
oxybutynin	Ditropan XL	Do not chew or crush. Slow release. Note: Immediate-release tablets may be crushed and can be used as an alternative.
oxycodone	Oxycontin	Do not chew or crush. Note: Crushing, chewing, or dissolving tablets can cause rapid release and absorption of a potentially fatal dose.
paliperidone	Invega	Do not chew or crush. Slow release.
pantoprazole	Protonix	Do not chew or crush. Slow release.
paroxetine	Paxil	Do not chew or crush. Film-coated.
pentoxifylline	Trental	Do not chew or crush. Extended release.
potassium chloride	K-Dur	Do not chew or crush. Note: Use alternative formulation.
propafenone	Rythmol SR	Do not chew or crush. Slow release.
propranolol	Inderal LA	Do not chew or crush. Slow release.
quetiapine	Seroquel XR	Do not chew or crush. Slow release.
raltegravir	Isentress	Do not chew or crush film coated 400mg tablets. If needed, chewable tablets are an alternative.
ranolazine	Ranexa	Do not chew or crush. Slow release.
rilpivirine	Edurant	Do not crush
ritonavir	Norvir	Do not chew or crush. Note: Crushing tablets has resulted in decreased bioavailability of drug.

sacubitril/valsartan	Entresto	Do not open or split. Not enough guidance available. If required, oral suspension may be compounded using 49/51mg tablets (see Micromedex).
tacrolimus	Prograf	Do not open or crush capsules for NG/G tube administration. Note: Pharmacy to compound suspension using oral capsules.
tamsulosin	Flomax	Do not chew or crush. Slow release.*
temozolomide	Temodar	Do not open capsules, crush or chew. Note: avoid inhalation or contact with the skin or mucous membranes.
theophylline	Theo-24	Do not chew or crush. Extended release.
tolterodine	Detrol LA	Do not chew or crush. Extended release.
valganciclovir	Valcyte	Do not chew or crush. Teratogenic and irritant potential
venlafaxine	Effexor XR	Do not chew or crush. Slow release. Note: Utilize immediate-release tablet formulation if needed.
zidovudine	Retrovir	Do not crush. Liquid formulation available
ziprasidone	Geodon caps	Do not open, crush or chew capsules.

Table 2: Can be crushed, but handling precaution required.

Generic (Active Ingredient)	Drug Brand Name	Administration comments
divalproex capsule	Depakote Sprinkles	Sprinkle capsules may be opened, and contents mixed with teaspoonful of soft food; swallow the drug/food mixture immediately (do not chew). Use double gloves.
finasteride	Proscar	Crush/disperse only using a closed system using PPE. Women who are or may become pregnant should not handle crushed or broken tablets due to teratogenic potential.
hydroxyurea	Hydrea	Exposure to the powder may cause serious skin toxicities, health care workers should wear double gloves to administer.
raloxifene	Evista	Due to taste and teratogenic potential, healthcare workers should wear double gloves and a protective gown.

Table 3: Can be Crushed, but taste may limit tolerability.

Generic (Active Ingredient)	Drug Brand name	Administration comments
brivaracetam	Briviact	Tablets may be crushed, but taste may limit tolerability.
cefuroxime	Ceftin	Tablets may be dispersed in water (20ml)
indinavir	Crixivan	Due to taste, capsule may be opened and mixed with fruit puree
levetiracetam	Keppra	Do not chew or crush as taste may limit tolerability. Utilize liquid formulation as an alternative.

Table 4: Can be Administered with Special Instructions

Generic (Active Ingredient)	Drug Brand name	Administration comments
amphetamine-salt	Adderall XR	Do not chew or crush. Capsule may be opened, and the contents can be mixed with soft food such as applesauce, contents may be administered via NG tube using an appropriate fluid provided entire contents are washed down the tube.
artemether-Lumefantrine	Coartem	May be crushed and mixed with one-two teaspoons of water.

atazanavir	Reyataz	Capsule may be opened and contents mixed with 4oz of applesauce. Must be administered within 5 minutes after a light meal.
bictegravir,emtricitabine,tenofovir alafenamide	Biktarvy	May dissolve tablet in water or orange juice.
carvedilol	Coreg CR	Do not chew or crush. May add contents of capsule to chilled applesauce and consume immediately.
Cenobamate	Xcopri	Crushed tablet may be mixed with 25ml of water for NG tube administration.
Cinacalcet	Sensipar	May be crushed. Crush each 30mg tablet with 10ml of water for administration.
doxycycline	Doryx	May be crushed, but risk of gastritis exists. May be compounded into solution if needed.
duloxetine	Cymbalta	Do not crush. Capsule may be opened, and contents taken without crushing or chewing; soft foods such as applesauce, contents may be administered via NG tube using an appropriate fluid provided entire contents are washed down the tube.
Emtricitabine/ tenofovir disoproxil	Truvada	Tablet may be crushed and disintegrated in water.
ergocalciferol	Drisdol	Do not chew or crush. Liquid-filled. Capsule may be opened, and the liquid contents removed for administration.
esomeprazole	Nexium	Capsule may be opened and the contents taken without chewing or crushing; soft food such as apple sauce or pudding may facilitate administration; contents may generally be administered via NG tube using an appropriate fluid provided entire contents are washed down the tube.
etravirine	Intelence	Tablet should be swallowed whole and not crushed; tablet may be dispersed in water and given by mouth or feeding tube.
gabapentin	Neurontin	May open capsule and mix with 4oz applesauce or 120ml orange juice.
hydroxychloroquine	Plaquenil	Tablets may be crushed and mixed with small amount of applesauce, or an extemporaneous suspension may be compounded
imatinib	Gleevec	Do not chew or crush. Mucous membrane irritant. Tablet may be dissolved in water or apple juice.
lansoprazole	Prevacid Solu-tab	Do not chew or crush. Orally disintegrating. Dissolve in 10ml water only and dispense via dosing syringe or NG tube.
memantine	Namenda XR	Do not chew or crush. Capsule can be opened and sprinkled on apple sauce.
mesalamine	Pentasa	Capsules may be opened and the entire contents sprinkles onto applesauce or yogurt. Contents should be consumed immediately, do not chew or crush.

pancrelipase	Creon	Do not chew or crush. Capsule may be opened, and contents taken without crushing or chewing; soft foods such as applesauce, contents may be administered via NG tube using an appropriate fluid provided entire contents are washed down the tube.
potassium chloride	Micro K/ K Dur	Do not chew or crush, Open capsule and mix in at least 2 ounces of liquid.
topiramate	Topamax, Topamax Sprinkles	Do not chew or crush. Capsule may be opened, and the contents taken without chewing or crushing; soft food such as applesauce or pudding may facilitate administration; contents may be generally be administered via NG tube using an appropriate fluid provided entire contents are washed down the tube.

***Notes:**

Diltiazem IR: ISMP states not to split, chew or crush. Conflicting information exists when reviewing Micromedex and other tertiary sites. Teva pharmaceutical labeling specifically states their product CAN be crushed or chewed and therefore, diltiazem IR has been removed from this list as of 2022. Note that diltiazem IR 30mg tabs can't be split in half.

Tamsulosin: Studies in pediatrics allowed opening capsules and mixing with acidic food (apple sauce, juice) without chewing or crushing the contents. Lack of studies exist for opening capsules for administration via tubes. If no therapeutic alternative available for patient with G tube, may consider administration by opening capsule and mixing with juice. Do not use with small bore tubes. Monitor patient for postural hypotension, dizziness, vertigo.

References:**1. HIV Medication Crushability Reference:**

- a. Immunodeficiency Clinic. [LIQUID DRUG FORMULATIONS \(hivclinic.ca\)](https://www.hivclinic.ca/liquid-drug-formulations)
- b. Zamora, J. A Complete Guide on Crushing Tablets and/or Opening Capsules of Human Immunodeficiency Virus Medications. Infectious Disease in Clinical Practice. 2017.
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2. Institute for Safe Medication Practices. Oral Dosage Forms That Should Not Be Crushed. 2020.
3. Uttaro E, Zhao F, Schweighardt A. Filling the Gaps on the Institute for Safe Medication Practices (ISMP) Do Not Crush List for Immediate-release Products. International Journal of Pharmaceutical Compounding. 2021; 25(5):364-71.